MILITARY SPECIFICATION

ENGINE, DIESEL: 6 CYLINDER, IN-LINE,
(REBUILD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the rebuilt Model D33T Diesel 6 cylinder, in-line, liquid-cooled, 4 stroke-cycle, turbo-supercharged, internal-combustion engine as modified for use in 8 ton GCER military vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

P-C-437	-	Cleaning Compound, High Pressure (Stream) Cleaner.
VV-F-800	-	Fuel Oil, Diesel.

Military

MIL-P-514	-	Plates; Identification, Instruction and Marking, Blank.
MIL-L-2104	-	Lubricating Oil, Internal-Combustion Engine, Tactical Service.
MIL-L-10295	-	Lubricating Oil, Internal-Combustion Engine Sub-Zero.
MIL-B-11188	-	Batteries, Storage: Lead-Acid.
MIL-C-11755	-	Compound Antifreeze, Arctic-Type.
MIL-S-13856	-	Electrical Components for Automotive Vechicles; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for.
MIL-A-46153	-	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

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MIL-T-62248(AT)	-	Truck, Tank: Fuel Servicing 2500 Gallon, 4x4, M559 (Rebuilt).
MIL-T-62249(AT)	-	Truck, Wrecker: 10 Ton, 4x4, M553 (Rebuilt).
MIL-T-62250(AT)	-	Truck Cargo: 8 Ton, 4x4, M520 and M877 (Rebuilt).
MIL-T-62252(AT)	-	Transmission, Power Shift, Commercial Modified (Rebuilt).
MIL-C-62253(AT)	-	Crane: Material Handling.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by attributes.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-193	-	Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461A Notice 4	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-721	-	Definitions of Effectiveness Terms for Reliability Maintainability, Human Factors, and Safety.

DRAWINGS

11681062	-	Power Plant Assembly.
11614457	-	Transmission Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Control of Air Pollution From New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should reference "Federal Register Volume 35, Number 108, Part II, dated June 4, 1968" and should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Accessories and equipment. All electrical accessories and equipment shall conform to the applicable requirements of MIL-E-13856, and specifications shown on the accessory drawing. All accessories and equipment installed on the engine shall be properly adjusted as required for vehicle operation.

3.2 Assembly. The rebuilt engine assembly including accessories shall be in accordance with drawing 11681062. The engine shall interface and be

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compatible with new or rebuilt transmission 11614457 procured under MIL-T-62252. The rebuilt engine shall provide the necessary performance and durability to meet the requirements of this specification and vehicle specifications. The engine shall be identified in accordance with MIL-STD-130. Spare parts, assemblies, and subassemblies shall be identified in accordance with MIL-STD-130 when specified on drawings. Definitions contained in MIL-STD-721 shall apply.

3.2.1 Interchangeability of parts. Component assemblies and parts of the engine, shall be so constructed that they may be installed, replaced, and adjusted without requiring modification among other engines provided under the same part number. If some parts must be furnished in matched sets, the rebuilder shall provide a list giving part number, part nomenclature, and tolerance range of sets involved. This list must be approved by the government prior to shipment of any engines.

3.2.2 Qualified products. When assemblies and parts requiring qualification are specified, the rebuilder shall be responsible for using items that are listed or approved on the qualified products list (QPL). The rebuilders inspection records shall list all QPL items by part or drawing number, name, and supplier and date of qualification.

3.3 Reliability, durability, maintainability, and endurance requirements.

3.3.1 Reliability (Minimum). The rebuilt engine shall have a 1000 hour mean time between failure (MTBF) when installed in either the M520, M553, M559 or M877 vehicles. Failure is defined as any primary malfunction which causes the rebuilt engine to cease operation, produces a safety hazard, or will contribute to engine damage if operation is continued.

3.3.2 Durability (Minimum). Rebuilt engine replacements in vehicles shall be used to measure durability and shall have probabilities of not requiring replacement as follows:

a. M520 68% in 20,000 miles of operation.
b. M553 82% in 10,000 miles of operation.
c. M559 77% in 14,500 miles of operation.
d. M877 68% in 20,000 miles of operation.

3.3.3 Maintainability. When maintenance procedures shown in the technical manuals are followed and the rebuilt engine is operated in the vehicles, as defined in the vehicle mission statements, the demonstrated maintainability shall be as follows:

(a) Mean Operator Preventive Maintenance Time shall not exceed .25 manhours per mission. This time shall not be included in organizational preventive maintenance time.
(b) Maximum Operator Corrective Maintenance Time shall not exceed 1.00 manhours in vehicle per mission without being classified as a mission failure.

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(c) The ratio of total corrective and organizational preventive maintenance manhours to operating hours shall not exceed 0.07.
(d) The ratio of total organizational preventive maintenance manhours to total operating hours shall not exeed .02.
(e) The ratio of total corrective maintenance manhours to operating hours shall not exceed 0.05.
(f) Mean manhours to perform a corrective maintenance action shall not exceed 4.00.
(g) The Mean Time Between Corrective Maintenance Actions shall not be less than 100 operating hours.
(h) Inherent Availability (A_i) shall not be less than 0.86.

$$A_i = \frac{MTBMA}{MTBMA + MTTR}$$

NOTE: MTBMA and MTTR are expressed in hours.

3.3.4 Endurance run. The bare rebuilt engine (see 6.3) shall be subjected to an endurance run as specified (see 4.3.1). At the conclusion of the run, the engine shall conform to all the performance requirements of this specification with the exception that the power and torque shall be not less than 95% of initial readings without readjustment of the injection system. In addition, there shall be no evidence of abnormal wear, scuffing of galling of cylinder bores, pistons, piston rings, piston pins, bearing journals, gear teeth, cam surfaces, valves, tappets or other parts, burning of valves or pistons, sticking of valves or piston rings, leakage of gaskets or seals which may result in failure of a component or other manfunction.

3.3.4.1 Initial rebuilt engine(s). The engine(s) selected shall be fully representative of engines proposed to be furnished and shall be submitted for inspection to determine conformance to the requirements of this specification.

3.3.5 Performance requirements.

3.3.5.1 Speed range. The engine shall operate under all loads and conditions as specified herein throughout the entire speed range. The engine shall maintain an idle speed of 700 to 800 rpm when using fuel specified herein.

3.3.5.1.1 Governor. The governor shall limit the maximum engine speed as follows:

Full load operation	2175 to 2225 rpm
No load operation	2330 to 2380 rpm

The engine speed with no load on the engine shall stabilize within 3 seconds after full-throttle position is reached. The governor shall have been set and sealed prior to shipment.

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3.3.5.2 Brake horsepower. At the completion of the rebuild break-in run, the bare engine (see 6.3) at the full-rack (throttle) setting shall develop 213 minimum gross brake horsepower (see 6.4) at 2200 rpm when using fuel specified herein, air density of 0.070 pounds per cubic foot, a fuel temperature of 90°F, and fuel at specific gravity of 0.853 at 60°F.

3.3.5.3 Torque. At the completion of the rebuild break-in run and with the full-rack setting specified herein, the bare engine (see 6.3) shall develop at the engine flywheel a minimum observed torque of 578 lb ft at 1550 rpm when using fuel specified herein.

3.3.5.3.1 Correction factors. Engine rating can be corrected as follows: 1.0 brake horsepower per each 1 change of API gravity.

3.3.5.4 Exhaust smoke density. The maximum exhaust smoke density at full rack position, through the speed range of 1550 to 2200 rpm when measured within three feet of the turbocharger outlet, shall not exceed a No. 3 visual or a No. 6 meter smoke condition as defined in 6.5 when using fuels specified herein.

3.3.5.4.1 Air pollution control. All vehicles shall comply with the Department of Health, Education, and Welfare Regulations governing control of air pollution from new motor vehicles and new motor vehicle engines in effect on the date of award of contract.

3.3.5.5 Fuel. The engine shall be capable of full performance when using fuel conforming to Grade DF-2 of VV-F-800. Using the specified fuels, the engine shall perform as specified with fuel inlet temperatures of up to 180°F measured at inlet to injection jump.

3.3.5.5.1 Fuel consumption. The fuel consumption in lbs per hour of a bare engine which meets the requirements of 3.3.5.2 and 3.3.5.3 (see 6.3) shall be not greater than the following when operating at full load, and full-rack setting and using fuel specified herein.

ENGINE - RPM	FUEL: POUND PER HOUR
1550	85.0
2200	100.0

3.3.5.6 Oil. Engine shall perform as specified herein using lubricating oil in accordance with the seasonal requirements of MIL-L-2104 from -10 to + 125°F, and MIL-L-10295 from -65 to + 10°F. The manufacturer may use oil of his own selection during engine build-up and break-in run.

3.3.5.6.1 Oil consumption. The engine shall consume not more than 0.0035 lb of lubricating oil per observed brake horsepower per hour, when operated throughout the endurance run, with oil sump temperatures of 230 to 250°F, using oil as specified herein. During vehicle operation with full pay load over hard-surfaced roads, the oil consumption shall be not more than 1 quart per 100 miles.

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3.3.5.6.2 Oil pressure. Under all operating conditions specified herein, including oil at the add oil mark and with the oil sump temperature of not less than 180°F, gallery oil pressure shall not be more than 80 pounds per square inch (psig) and not less than 32 psig when the engine is operating at 2200 rpm, using oil as specified herein and shall not be less than 5 psig when engine is idling as specified herein.

3.3.5.7 Supercharging. The turbo supercharger shall develop a minimum compressor discharge pressure of 50 in. hg. absolute when engine is operating at 2200 rpm with full load and using fuel conforming to VV-F-800.

3.3.5.7.1 Air induction restriction. The engine shall perform as specified (see 3.3.5.1) with inlet restriction of 10 to 15 inches of water measured at inlet to turbo supercharger.

3.3.5.7.2 Exhaust back pressure. The engine shall perform as specified (see 3.3.5.1) with an exhaust back pressure of 1.5 to 2.0 inches of hg. measured at the outlet of the turbo supercharger.

3.3.5.8 Operating temperatures. The engine oil and coolant shall not exceed the following temperatures when operating at full load throughout the speed range of the engine under any of the conditions specified herein or vehicle operation.

(a) Oil Sump	250°F.	(measured at the drain plug)
(b) Coolant	230°F.	(measured at the engine outlet)

3.3.5.9 Cooling system. With a water pump inlet restriction of not more than 1" Hg vacuum and engine outlet restriction pressure of not more than 20.4" Hg (10 PSIG), the outlet thermostat in wide open position, and an inlet temperature of 195° to 210°F, the coolant temperature rise through the engine block and cylinder heads shall not exceed 12°F, when operating at 213 HP and 2200 RPM in the vehicle. For bare engine testing, an inlet temperature of 165° to 175°F may be used, and the coolant temperature rise through the engine block, cylinder heads, and oil cooler shall not exceed 20°F. This requirement shall be met with all mixtures of water containing from 0 to 50% antifreeze compound conforming to MIL-C-11755.

3.3.5.9.1 Coolant. The engine shall perform as specified herein using water and antifreeze compound conforming to MIL-A-46133, arctic-type antifreeze compound conforming to MIL-C-11755, or water corrosion inhibitor compound conforming to O-I-490, without overheating and damage to engine.

3.3.5.10 Environmental requirements.

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3.3.5.10.1 Starting. The engine shall start within 2 minutes after recommended use of starting aids under any of the following conditions or possible combination:

a. With integral cold starting aid glow plugs, and without external aids after being cold-soaked for a 24 hour period, and without benefit of solar radiation to an ambient temperature of -25°F. Electrical power shall be supplied by four fully charged 6 TN batteries conforming to MIL-B-11188 that were cold-soaked for 24 hrs and with the wiring system same as used on the vehicle.

b. Without external aids, with cooling fan installed, and a fuel pressure of 5 psi at the inlet to the fuel transfer pump, and when operated at the following temperatures with exposure to maximum solar radiation:

Ambient air temperature	125°F
Intake air temperature (at air cleaner inlet)	125°F
Coolant, engine out temperature	230°F
Oil sump temperature	At maximum temperature attainable up to 250°F

c. With the transmission and all accessories when installed in the vehicle.

d. Humidity conditions. Under relative humidity conditions as low as 5 percent at temperature of 125°F. and as high as 100 percent at all temperatures from -25 to 85°F.

3.3.5.10.2 Elevation conditions. The power output of the engine shall remain unchanged at any elevation from sea level to 10,000 feet at ambient temperatures not greater than 60°F.

3.3.5.11 Grades and slopes. The engine shall start and demonstrate the performance characteristics specified herein when operated. The engine shall operate with full, and add oil, oil level conditions in the sump, on longitudinal grades up to 60 percent or lateral slopes up to 30 percent in all directions and combinations. As a result of said operation, no evidence of faulty lubrication, cooling, fuel supply, leakage, or other malfunction shall be found.

3.3.5.12 Steam and water jet cleaning. The engine and all its components shall withstand cleaning with high pressure steam and cleaner conforming to P-C-437 and water jet, without deterioration of seals or hose, water leakage past seals or gaskets, or other defects. Paint removal shall not be considered a deficiency. Steam and water jet cleaning is defined as follows:

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The jet is applied perpendicular to the surface being cleaned at a distance of not more than 1 ft. from the surface for steam cleaning, and not more than 5 feet from the surface for water jet cleaning, and cleaned at a rate of 1 square foot per minute. The jet pressure shall be not less than 100 psi, and not more than 110 psi.

3.4 Rebuilt engines. All engines procured under the rebuild contract shall meet all the operational and performance requirements specified herein.

3.4.1 Rebuild break-in run. Each engine shall be given a break-in run by the rebuilder. After successful completion of this run, the engine with no further break-in, shall be ready for use in the Goer vehicles and meet all performance requirements specified herein.

3.4.2 Fifty hour run. (Control Test) an engine selected from rebuilt stock shall be submitted to a 50 hour run using fuel conforming to Grade DF 2 of VV-F-800.

3.5 Exterior surface treatment. Pretreatment prior to painting shall be according to rebuilder's standard commercial practice and shall be painted in accordance with the applicable provisions of MIL-STD-193.

3.6 Nameplate. Unless otherwise specified (see 6.2) on the applicable drawings, the nameplate, data and instruction plates shall conform to the applicable provisions of MIL-P-514.

3.7 Radio, interference suppression. The engine shall meet the tactical suppression requirements of MIL-STD-461A, Notice 4.

3.8 Workmanship. The engine assembly shall be rebuilt and finished thoroughly in a workmanlike manner, particular attention shall be paid to thoroughness of painting, plating, bolted on assemblages, welding and brazing. Electric, fuel and lubrication lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. All parts and assemblages shall be free from burrs, sharp edges or any condition which may present a safety hazard to operating and maintenance personnel.

3.9 Leakage. During dynamometer operation or when installed in vehicle, the engine shall show no evidence of leakage greater than seep as defined in 6.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the rebuilder is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the rebuilder may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Rebuilder's inspection or quality control system. Unless otherwise specified by the procuring activity, the rebuilder shall provide and maintain an effective inspection or quality assurance system acceptable to the Government covering the supplies under the contract. A current, written description of the system shall be submitted to the contracting officer prior to initiation of production. The rebuilder shall notify the Government of any change to the written procedure subsequent to submission of procedure to the contracting officer.

4.1.2 Government verification. All quality assurance operations performed by the rebuilder will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the rebuilder. Failure of the rebuilder to promptly correct defects discovered shall be cause for suspension of acceptance until corrective action has been taken, or until conformance of product to specified criteria has been demonstrated.

4.1.3 Materials and qualified products. The rebuilder shall provide the Government with evidence that materials used are as specified in the applicable specifications and drawings (see 3.1, 3.2, 3.2.1, and 3.2.2).

4.1.4 Break-in. Every engine shall be operated as specified (see 3.4.1) and any unsatisfactory engine operation or condition which requires correction shall be corrected by the rebuilder before the engine is presented for acceptance or additional testing.

4.1.5 Log sheet. The rebuilder shall maintain a log sheet of each engine run during all tests specified herein. The form of the log sheet and number of legible copies to be furnished to the Government shall be as approved by the procuring activity.

4.2 Inspection provisions.

4.2.1 Lot size. An inspection lot shall consist of all engines of one type from one day's rebuild, submitted at one time for acceptance inspection.

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4.2.1.1 Sampling for inspection. For the purpose of visual, dimensional and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the rebuilder, the rebuilder shall 100% inspect the first 20 engines to establish a process average (see 6.9) to allow normal sampling in accordance with MIL-STD-105.

4.2.1.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the engines or conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table I. The following constitutes a part of the classification of defects (see 6.6 and 6.10):

(a) Any drip constitutes a major defect when the engine has been standing idle and the components are at ambient temperature.
(b) Any droplet that occurs at a static fit metal to metal or gasket combination shall constitute a major defect.

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Table I. Classification of quality defects

Defect No.	Defects	Method of Inspection
Major	A.Q.L. = 25 Defects/100 Units	
101	Valve tappet clearance (each valve): intake and exhaust valves improperly adjusted	Visual and functional
102	Fuel, oil and coolant leaks: leakage (see 6.6)	Visual
103	Governor: malfunction; not sealed	Visual and functional
104	Torque on cylinder head bolts, intake and exhaust manifold flange bolts: improper torque	Visual and functional
105	Crankshaft: excessive or restrictive end play	Visual and functional
106	Fuel system components: malfunction; damage; leaks (see 6.6)	Visual and functional
107	Fuel injection system timing and components: malfunction; improper adjustment	Visual and functional
Minor	A.Q.L. = 150 Defects/100 Units	
201	Fuel lines: damaged; rubbing; improper support	Visual
202	Injection pump support bracket: improperly installed	Visual
203	Linkage: improper adjusted	Visual
204	Minor assemblies: omitted	Visual
205	Engine: improper adjustment or installation of components	Visual and functional
206	Fuel lines, hose, vents, shut-off valve: improperly assembled or installed	Visual
207	Painting: spots missed; sags or runs	Visual

4.2.1.3 Unclassified defects. All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day period will be added to the minor classification with no increase to A.Q.L.'s.

4.2.1.4 Recurring major deficiencies. A major deficiency (see 6.10) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

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4.2.2 Examination failure. If an engine fails to pass any acceptance examination specified herein, the government inspector shall stop acceptance of subsequent engines until evidence has been provided by the rebuilder that corrective action has been accomplished. Correction of any defect must be accomplished by the rebuilder at no cost to the Government on all engines accepted under the contract. Any defects found are considered prima facie evidence that engines already accepted are similarly defective unless evidence satisfactory to the contracting officer is furnished by the contractor that no defects exist.

4.2.2.3 Maintainability. The maintainability of the rebuilt vehicle as specified in 3.2.3 will be verified during rebuilt vehicle test.

4.3 Initial rebuilt test. One of the first ten production engines produced under contract will be selected by the Government and subjected to a 320 hour endurance run as specified in table II. This test shall be conducted by the Government at a place designated by the Government to determine conformance to 3.3.4.1 and 3.3.4. With the engine, the rebuilder shall furnish a certified graph with performance curves showing horsepower, torque, and specific fuel consumption for the bare engine. The rebuilder shall support the dynamometer endurance test and the initial rebuilt vehicle tests by providing any components necessary for completion of such test. Delay in test completion caused by engine breakdown or failure of the rebuilder to adequately support the engine with parts during the tests shall not be the basis for adjustment of contract delivery schedule or contract price. In addition, the initial rebuilt test engine must be examined for all defects of table I and all tests as required by table III.

4.3.a In vehicle test. To determine conformance to 3.3.1 through 3.3.3 engines will be installed in test vehicles (3.3.3) (quantities as specified in the contract) and shall be subjected to mileage specified for each vehicle type. Those tests (3.3) shall be conducted in conjunction with the vehicle initial test.

4.3.b Failure. Any engine test failure found during the above test shall be reason for refusal by the Government to accept subsequent engines until evidence has been provided by the supplier that corrective action has been taken. Any test failures found during or as a result of these tests shall be prima facie evidence that all engines already accepted prior to completion of test are similarly defective unless evidence satisfactory to the Government is furnished by the rebuilder that they are not similarly defective. Such test failures on all engines shall be corrected by the rebuilder at no cost to the Government.

4.3.1 Endurance tests. The bare engine (see 6.3) shall be operated in accordance with table II to determine conformance to 3.3.4.

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Table II. Endurance five hour schedule

Period	% Rated Speed	% Load	Time, Hrs
1	Low Idle	0	1/2
2	Max. Torque	100	1
3	100	0	1/2
4	75	85	1
5	100	50	1/2
6	100	100	1
7	50	25	1/2
		TOTAL	5 hours

4.3.1.1 Performance tests.

4.3.1.1.1 Speed range and governor tests. To determine conformance to 3.3.5. 1 and 3.3.5.1.1, the speed range and governor shall be checked for functional requirements.

4.3.1.1.2 Brake horsepower and torque test. To determine conformance to 3.3.5.2 and 3.3.5.3, the engine shall be operated, as specified, brake horsepower and torque checked for functional requirements.

4.3.1.1.3 Exhaust smoke density test. To determine conformance to 3.3.5.4, with the engine operating, as specified, the smoke shall be checked for density.

4.3.1.1.4 Fuel consumption test. To determine conformance to 3.3.5.5.1, with the engine operating as specified; the fuel consumption shall not exceed that specified.

4.3.1.1.5 Oil consumption and pressure tests. To determine conformance to 3.3.5.6.1 and 3.3.5.6.2, with the engine operating as specified; the oil consumption and oil pressure shall be not more than that specified.

4.3.1.1.6 Supercharging, air restriction, and exhaust back pressure test. To determine conformance to 3.3.5.7 through 3.3.5.7.2, with the turbo supercharger operating as specified, the discharge pressure, air induction restriction, and exhaust back pressure shall be measured.

4.3.1.1.7 Operating temperatures test. To determine conformance to 3.3.5.8, with engine operating as specified; the oil and coolant shall be measured at locations indicated and temperatures checked.

4.3.1.1.8 Cooling system test. To determine conformance to 3.3.5.9, the cooling system with specified restrictions shall not cause the coolant temperature rise through the engine more than specified.

4.3.1.1.9 Grades and slope test . To determine conformance to 3.3.5.11, the initial rebuilt engine shall be subjected to this test.

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4.3.1.1.10 Steam and jet cleaning test. To determine conformance to 3.3.5.12, the initial rebuilt engine shall be subjected to cleaning as specified. The seals, hose, and gaskets shall be checked for damage and leaks.

4.3.1.1.11 Air pollution test. To determine conformance to 3.3.5.4.1, the initial rebuilt engine shall be tested for compliance to specific standards.

4.3.1.1.12 Radio interference suppression test. The engine, as delivered, shall have met the requirements of specified standard.

4.3.1.2 Environmental tests. To determine conformance to 3.3.5.10, the engine shall be subjected to specified environment conditions verified.

4.3.1.2.1 Starting test. To determine conformance to 3.3.5.10.1, the engine shall be subjected to a starting test under specified conditions.

4.3.1.2.2 Elevation test. To determine conformance to 3.3.5.10.2, the engine power output shall be verified at specified altitudes. When actual altitude or altitude chamber is not available, the conditions shall be simulated by imposing restriction on inlet air and suction on engine exhaust.

4.3.1.2.3 Leakage test. To determine conformance to 3.9, with engine operating or installed in vehicle, leakage shall be not more than a weep or seep as indicated in 6.6.

4.3.1.3 Endurance test. The endurance test shall consist of a total of 320 hours divided into four periods of 80 hours each. Each 320 hour period shall consist of 16, five hour endurance schedules (see table II). Procedures defined in 6.8 will be utilized.

4.3.1.3.1 Test hours. Any period of the schedule terminated due to equipment malfunction or minor test specimen failure must be repeated when time elapsed is less than half. The voided part of the period will not be accrued as endurance time.

4.3.1.3.2 Power check tests. To insure engine compliance to performance requirements lists below, the engine shall be subjected to power checks before start of the endurance run and thereafter every 80 hours of endurance time:

Speed Range	(see 3.3.5.1)
Brake Horsepower	(see 3.3.5.2)
Torque	(see 3.3.5.3)
Exhaust Smoke Density	(see 3.3.5.4)
Fuel Consumption	(see 3.3.5.5.1)
Oil Pressure	(see 3.3.5.6.2)
Supercharger Pressure	(see 3.3.5.7)

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The power checks will be performed with the throttle set in the wide open position (full load) and engine speeds of 1550, 1700, 1800, 1900, 2000, 2100 and 2200 RPM in both ascending and descending order. At each engine setting, the engine should be run for a sufficient time (approximately 20 minutes) for stabilization. In addition, following the pre-endurance power check, part load data will be obtained at each of the speeds above and 85%, 70%, 50% and 25% of the full load for each speed.

4.3.1.3.3 Temperatures. The endurance test, including power checks, will be performed at an ambient cell temperature or 85 ± 5°F, except for the last 80 hour period which will be performed at an ambient cells temperature of 125 ± 5°F.

4.3.1.4 Tear down inspection. Prior to tear down inspection, a compression check shall be made on all cylinders. Variation of compression between cylinders shall be not more than 50 psi at 400 rpm's. Each engine subjected to the tests shall be disassembled to the extent required to perform the following inspections:

a. Oil contamination.
b. Dirt, chips or foreign matter in the entine block, oil pan, filters, and accessory drive case.
c. Main bearings bolt torque.
d. Connecting rod bolt torque.
e. Cylinder bore scuffing, scoring, and galling.
f. Piston scuffing, scoring, and burning.
g. Excessive wear.

4.3.1.5 Endurance failure. Failure of an engine to conform to any of the endurance requirements specified or any major defect in workmanship or material found as a result of the test shall be cause for rejection, and for refusal to continue acceptance of rebuilt engines until objective evidence has been provided by the rebuilder that corrective action has been taken to eliminate the defect. Correction of any defect must be accomplished by the rebuilder at no cost to the Government on engines accepted under the contract. Any defects found as a result of test is considered prima facie evidence that engines already accepted are similarly defective unless evidence satisfactory to the contracting officer is furnished by the rebuilder that no defects exist.

4.3.1.5.1 A major failure. Is a failure to any component or part of the engine system that causes an engine stoppage or reduction of power in excess of 15 HP. All major failures shall be a basis for an endurance test termination. Subsequent retest will begin at 0 hours on the 320 hour schedule.

4.3.1.5.2 A minor failure. Is a defect leading to a reduction of power not exceeding 15 HP provided the defect can be rectified and 95% of full power restored within the scope of normal maintenance and adjustments permitted at

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the completion of the 80 hour period. Failure to attain 95% of full power after adjustment shall be a basis for termination. Subsequent retest will begin with 0 hours on the 320 hour schedule.

4.4 Acceptance tests. After successful completion of the break-in run, each engine shall be examined for defects in the acceptance tests listed in Table III. Oil sump examination shall be accomplished on the first ten (10) engines of any contract except overlapping contracts.

4.4.1 Acceptance test failures. If an engine fails to pass any examinations or tests specified herein, the Government shall stop acceptance until evidence has been provided that the corrective action has been taken on all engines.

Table III. Test schedule

Test No.	Name	Initial Rebuild	Acceptance	Control
4.1.4	Break-in run	X	X	
4.4	Acceptance	X		
4.3	Initial rebuilt (Endurance)	X		
4.3.1.1.1	Speed range & governor	X	X	X
4.3.1.1.2	Brake horsepower & torque	X	X	X
4.3.1.1.3	Exhaust smoke density	X	X	X
4.3.1.1.4	Fuel consumption	X	X	X
4.3.1.1.5	Oil consumption	X		X
4.3.1.1.5	Oil pressure	X	X	X
4.3.1.1.6	Supercharging	X		X
4.3.1.1.6	Air restriction	X		X
4.3.1.1.6	Exhaust back pressure	X		X
4.3.1.1.7	Operating temperatures	X		X
4.3.1.1.8	Cooling system	X		X
4.3.1.2	Environmental	X		
4.3.1.2.1	Cold starting	X		
4.3.1.2.2	Elevation	X		
4.3.1.2.3	Leakage	X	X	X
4.3.1.1.9	Grades & slope	X		
4.3.1.1.10	Steam & jet cleaning	X		

4.5 Fifty hour control test. The 50-hour control tests as referenced in 3.4.2 shall be conducted as specified herein for the purpose of determining the continuous quality of the engines. These tests are conducted at the rebuilder's plant.

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4.5.1 Selection of test sample. Engines shall be selected by the Government at the rate of one per month when rebuild is 100 unit per month or less and at the rate of two per month when rebuild is greater than 100 units per month. Engine shall be identified as to rebuild period.

4.5.2 Test. Each engine selected shall be examined for defects in table I and subjected to the tests in table IV.

Table IV. 50 hour control test

Period	Time (hrs)	Speed (rpm)	Percent of full rack load at indicated speed
1	Engine must have completed acceptance test)		
2	5	1600	50
3	5	1800	75
4	5	2000	75
5	5	2100	100
6	30	2200	100
7	(Performance as required, see table III.)		
8	(Disassembly & inspection)		
9	(Assembly)		
10	(Rerun acceptance test prior to shipment)		

4.5.3 Control test failure. Failure of engine selected for control tests to conform to the requirements specified or any major defect for workmanship or material found as a result of the test shall be cause for rejection and for refusal to accept subsequent engines. Also, defects found during or as a result of the test shall be prima facie evidence that engines accepted subsequent to the previous acceptable control test engine are similarly defective unless evidence satisfactory to the contracting officer is furnished by the rebuilder that they are not similarly defective. Defects on all engines shall be corrected by the rebuilder at no cost to the Government. Another engine, with corrective action implemented, shall be subjected to the control test.

4.5.4 Tear down inspection. Prior to tear down inspection, a compression check shall be made on all cylinders. Variation of compression shall be not more than 50 psi at 400 rpm's. Each engine subjected to the control tests shall be disassembled to the extent required to perform the following inspection:

(a) Oil contamination.
(b) Dirt, chips or foreign matter in the engine block, oil pan, filters and accessary drive case.
(c) Main bearings bolt torque, head bolt torque, intake and exhaust manifold flange bolts, and vibrator damper.

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(d) Connecting rod bolt torque.
(e) Cylinder bore scuffing, scoring, and galling.
(f) Piston scuffing, scoring, and burning.
(g) Excessive wear.

4.5.5 Corrections and reassembly. Should there be evidence of any condition of engine or parts requiring correction, such corrections shall be made by the rebuilder prior to reassembly of the engine and re-submission to the break-in run. In adition, the rebuilder shall submit evidence to the Government, that corrective action has been taken in production to eliminate any deficiency disclosed. Failure of the rebuilder to submit such evidence shall be cause for refusal by the Government to continue acceptance of subsequent engines. Any engine not requiring replacement parts shall be reassembled and subjected to the acceptance power check test only.

4.5.6 Test results and recorded data. The rebuilder's report furnished to the Government shall include all data resulting from the control test. This data shall be obtained under stabilized operating conditions and during the last half hour of periods 2 through 5. For period 6, the data shall be obtained during the last half hour of each 10 hour increments. Oil consumption shall be measured during the last 25 hours of operation.

4.6 Preparation for delivery. Inspection of preservation packaging, packing and marking shall be performed on each engine to determine conformance to 5.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preparation for delivery shall be in accordance with the technical requirements specified on the appropriate packaging data sheet (see 6.2).

6. NOTES

6.1 Intended use. The engines covered by this specification are intended for rebuild use, as spares, or as replacements for the M520, M553, M559 and M877 Goer vehicles.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Additional data for namelate, when required (see 3.6).
c. Quantity of initial rebuilt test engines required (see 4.2).
d. Selection of method and applicable level of preservation and packaging, and level of packing of the referenced specification (see 5.1).

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6.3 Bare engine. The term bare engine assembly, without compressor, is defined as an engine depicted on 11681061.

6.4 Brake horsepower. Brake horsepower is defined as the horsepower output of the bare engine at the flywheel.

6.5 Exhaust smoke density. Exhaust smoke density shall be determined with a Robert R Bosch Model EFAW 68 Smoke Meter or equal (as approved by responsible engineering activity) and Model DFAW 65 Sampling Pump or equal (as approved by responsible engineering activity). The following table may be used to visually define the degree of exhaust smoke density in lieu of the smoke meter:

Description of Exhaust Smoke	Classification
Clear	1
Haze	2
Light Gray	3
Medium Gray	4
Dark Gray to Black	5

NOTE: Observation of exhaust smoke should be made against a white background within 3 feet of the turbocharger outlet.

6.6 The following definitions shall be used for the classification of defects for leaks:

(a) Weep: - Any non-recurring evidence of fluid beyond the seal.
(b) Seep: - Any recurring evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: - Any recurring evidence of fluid beyond the seal that results in formation of a non-falling droplet.
(d) Drip: - Any recurring evidence of fluid beyond the seal where droplet forms and falls.

6.7 Typical vehicle life operational profile.

Model	Renewal Life	Total Missions
M520	20,000 miles (1000 hours)	200
M559	14,500 miles (1000 hours)	145
M553	10,000 miles (1000 hours)	200
M877	20,000 miles (1000 hours)	200

In actual troop use, this will vary due to environmental, terrains, and troop mission differences.

6.8 The following procedures are provided for conduct of endurance testing:

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6.8.1 Engine shut down. The engine test may be shut down after completion of any period in the five hour schedule. Planned shut down following test periods 2, 4, and 5 will include a cool down running of period No. 7. This cool down period will not be credited as endurance test time. This cool down period is not required after periods 1, 3, and 7.

6.8.2 Engine warm-up period. A one half hour warm-up period is required prior to continuation of endurance run following any shut down. During this one half hour period, the loads and speeds shall be progressively increased to the values required in period No. 5. Temperature stablization in period No. 5 is required. This warm-up period will not be credited as endurance time.

6.8.3 Data. The following data will be recorded during the last five minutes of each schedule periods and power check tests:

a. Engine speed (RPM).
b. Observed brake horsepower (OBHP) and/or torque (lb-ft).
c. Fuel flow (lbs/hour).
d. Test cell ambient temperature (^{o}F) and barometric pressure (in. Hg).
e. Engine oil gallery and sump temperatures (^{o}F) and pressure (PSIG-Gallery only).
f. Fuel secondary filter out and spill back temperatures (^{o}F) and pressures (PSIG).
g. Air cleaner inlet temperature (^{o}F).
h. Air intake manifold pressure (in. Hg).
i. Air cleaner restriction (in. water).
j. Coolant engine inlet and outlet temperatures (^{o}F) and pressures (PSIG).
k. Exhaust port temperatures (^{o}F).
l. Exhaust manifold pressure (in. Hg).
m. Crankcase pressure (in. water).
n. Fuel, transfer pump, inlet pressure (PSIG).
o. Oil consumption (lb/hr) will be calculated on basis of oil added in lbs accumulating to the engine as necessary during the endurance run.
p. Exhaust smoke density (power check only).

6.8.4 Warning and shut downs. To guard against overstress caused by equipment malfunction or human error, the following automatic warning/shutdown provisions will be incorporated in each test cell:

	Temperatures ^{o}F, maximum	WARNING	SHUT DOWN
a.	(1) Coolant, engine out	225	235
b.	(2) Oil, engine sump	245	255

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		WARNING	SHUT DOWN
b.	Pressure PSIG, minimum		
	(1) Oil engine gallery 600 - 1400 RPM	8	5
	(2) Oil engine gallery 1400 - 2200	18	10
c.	Engine speed	2380	2500

6.8.5 Servicing. Prior to the power check test, the lubricating oil shall be changed, all filters serviced, valve clearance and injector rack linkage checked and adjusted to applicable requirements.

6.9 Process average. Sampling may be initiated if the process average value of the first twenty engines inspected is less than the A.Q.L. specified in the classification of Defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of engines inspected}} \times 100$$

If the computed process average exceeds the specified A.Q.L. 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive engines is less than the specified A.Q.L.

6.10 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each engine is performed by the rebuilder not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the engine. The defects noted on the Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the rebuilder's representative to classify the deficiency in accordance with the classification of defects of the engine specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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